

6.6 (a) I don't know how to draw the configuration by hand, but it should be identical to that of Problem 5.32. From Problem 5.2 and 5.3, we know that the magnetic field inside the coil is along the axis of the coil, and is given by $\vec{B} = \pm \mu_0 N I \hat{\phi}$. The electric field is approximately

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 a^2} \hat{r} \text{ at the axis of the coil. Then,}$$

$$\vec{p}_{\text{field}} = \frac{1}{c} \int (\vec{E} \times \vec{B}) d^3x = \pm \frac{1}{c} \frac{Q}{4\pi\epsilon_0 a^2} \cdot N I \cdot A \hat{z} = \pm \frac{\mu_0 Q N I A}{4\pi a^2} \hat{z}$$

where the sign depends on the direction of the current flow.

To apply the approximation in Problem 6.5(b), notice that $\vec{m} = \pm N I A \hat{\phi}$ and we can arrive at the same result.